

MEDICAL INSURANCE COST ANALYSIS REPORT

Power BI Dashboard – Insights, Customer Segments & Cost Trends

Prepared by

Shivam Vishwakarma

1. Objective of the Report

This report presents an analysis of the Medical Insurance Cost Analysis Dashboard, which visualizes health insurance charges across customer demographics such as region, gender, number of children, age, BMI and smoking status. The objective is to summarize the key metrics visible in the dashboard, compare selected customer segments, and highlight the primary factors that appear to influence insurance costs.

The analysis is based on three views of the dashboard, captured with different filter combinations: an overall Northeast view (all genders, all children), a Northeast Male segment with 3 children, and a Southwest Male segment with 1 child.

2. Dashboard Overview

The dashboard tracks four headline KPIs — Total Insurance Cost, Total Customers, Average Insurance Cost and Maximum Insurance Cost — alongside six supporting visuals: Average Insurance Cost by Children, Insurance Cost by Region, Insurance Cost by Gender, Insurance Cost by Smoker, Average Insurance Cost by Age, and a BMI vs. Insurance Charges scatter plot. Interactive slicers for Region, Gender and Children allow the data to be filtered down to specific customer segments.

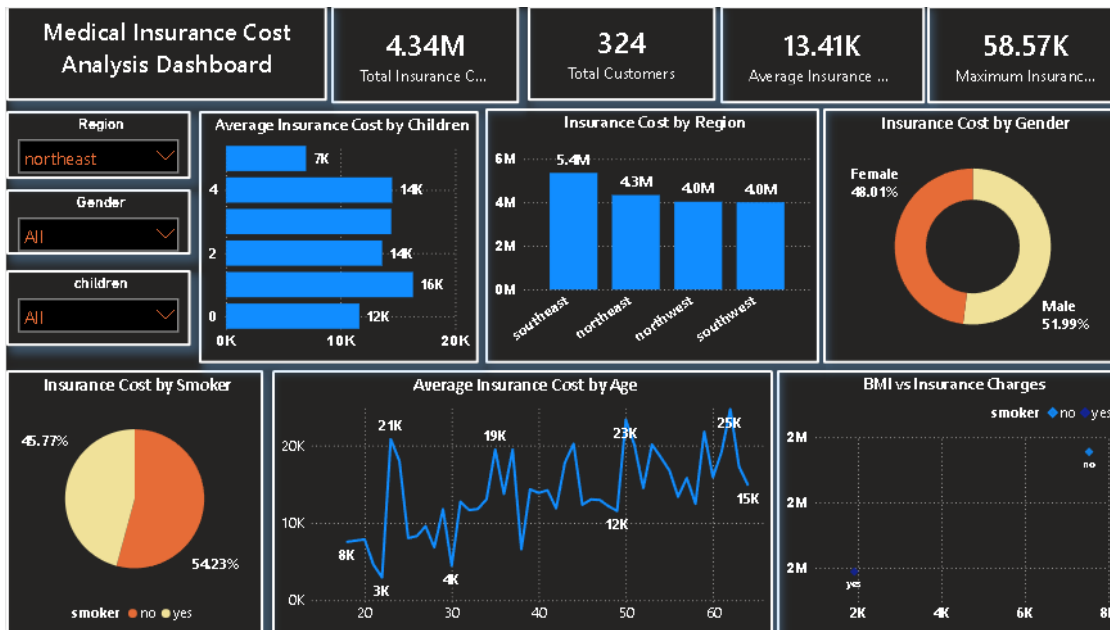


Figure 1: Overall dashboard view — Region = Northeast, Gender = All, Children = All

3. Key Metrics: Overall View (Northeast, All Customers)

- Total Customers: 324
- Total Insurance Cost: 4.34M
- Average Insurance Cost: 13.41K
- Maximum Insurance Cost: 58.57K
- Gender split is nearly even — Male 51.99%, Female 48.01%
- Smoker share is 54.23% (non-smoker) vs. 45.77% (smoker) in this view

Regionally, Southeast records the highest total insurance cost (5.4M), followed by Northeast (4.3M), while Northwest and Southwest are roughly equal at 4.0M each. Average cost by children shows a rising pattern, peaking at 16K for customers with 3 children before dropping for 4+ children. The Average Insurance Cost by Age chart is highly volatile, with visible peaks around ages 21, 46 and 61–63, suggesting cost is not driven by age alone but interacts with other factors such as smoking and BMI.

4. Segment 1 — Northeast, Male, 3 Children

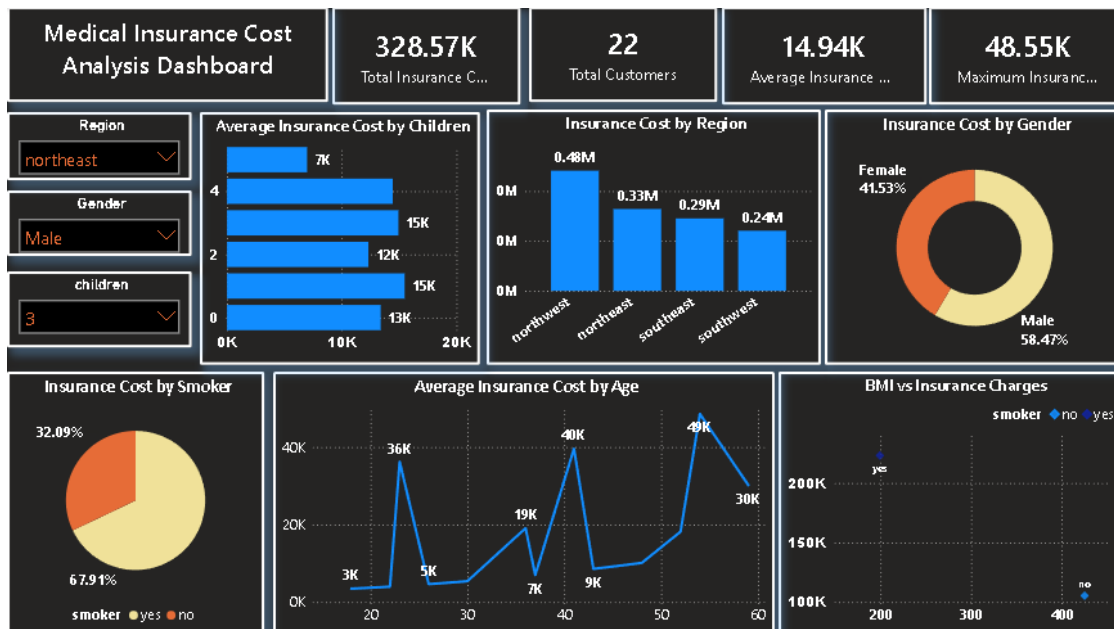


Figure 2: Filtered view — Region = Northeast, Gender = Male, Children = 3

- Total Customers: 22
- Total Insurance Cost: 328.57K
- Average Insurance Cost: 14.94K — higher than the overall Northeast average of 13.41K
- Maximum Insurance Cost: 48.55K
- Smoker share is only 32.09%, yet the average cost is still elevated

For this segment, Northwest region customers show the highest sub-total (0.48M) rather than Northeast, indicating the underlying dataset mixes region tags across visuals. The Average Insurance Cost by Age line is sharply spiked, with the cost jumping to 49K around age 55–58 before falling to 30K — a strong signal that a small number of high-cost individuals (likely smokers or high-BMI customers) are driving the average upward in the older age bracket.

5. Segment 2 — Southwest, Male, 1 Child

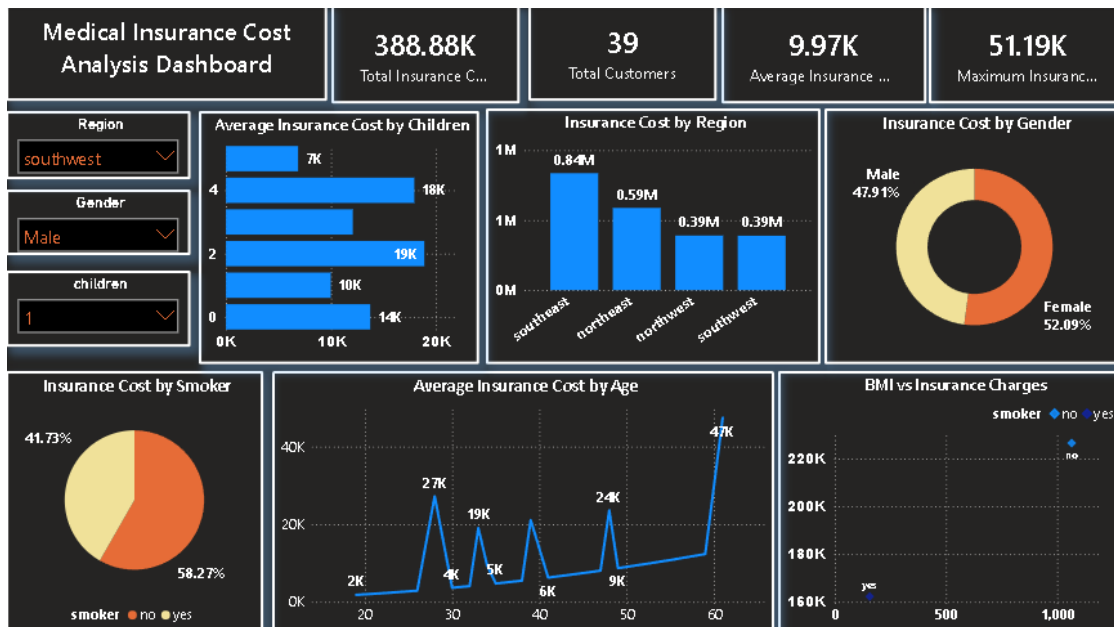


Figure 3: Filtered view — Region = Southwest, Gender = Male, Children = 1

- Total Customers: 39
- Total Insurance Cost: 388.88K
- Average Insurance Cost: 9.97K — the lowest of the three views
- Maximum Insurance Cost: 51.19K
- Smoker share is 58.27%, the highest among the three segments, yet the average cost is the lowest

This is a notable and slightly counter-intuitive finding: despite having the highest proportion of smokers, this segment shows the lowest average insurance cost. This suggests that within this specific segment, smoking status alone does not fully explain cost — age and BMI distribution likely play a larger role. The Average Insurance Cost by Age chart confirms this, showing costs largely under 10K until a sharp spike to 47K at age 60+, meaning most of the segment is low-cost with a small number of high-cost outliers near retirement age.

6. Comparative Summary

The table below places all three views side by side for quick comparison.

Metric	Overall (Northeast, All)	Northeast – Male (3 Children)	Southwest – Male (1 Child)
Total Customers	324	22	39
Total Insurance Cost	4.34M	328.57K	388.88K
Average Insurance Cost	13.41K	14.94K	9.97K
Maximum Insurance Cost	58.57K	48.55K	51.19K
Smoker Share	54.23%	32.09%	58.27%
Male / Female Split	51.99% / 48.01%	100% / 0%	100% / 0%

7. Key Insights

- Smoking status is a major cost driver overall, but its impact varies by segment — the Southwest/Male/1-child group shows the highest smoker share (58.27%) alongside the lowest average cost, indicating age and BMI moderate its effect.
- Number of children has a non-linear relationship with average cost — cost rises with children up to 3, then declines, likely reflecting family-size and age interactions rather than a direct cause.
- Age is not a smooth cost driver — every segment shows sharp spikes at specific ages (around the late 40s to early 60s) rather than a gradual increase, pointing to outlier high-cost customers rather than a uniform age effect.

- Regional cost totals shift depending on filters — Southeast leads in the overall view, while Northwest leads in the Male/3-children segment, showing that regional patterns are sensitive to demographic slicing.
- The BMI vs. Insurance Charges scatter plot consistently shows smoker customers positioned at higher charge levels than non-smokers at similar BMI values, reinforcing smoking as a compounding cost factor.

8. Recommendations

- Prioritize smoking-cessation and wellness programs for customers in high-smoker segments, since smoking consistently correlates with higher charges.
- Investigate the older-age cost spikes (55–63 range) further — a small number of high-cost customers appear to be skewing average costs and may warrant targeted risk assessment or premium review.
- Use segment-level analysis (region + gender + children) rather than single-variable analysis when setting premiums, as the data shows cost drivers interact rather than act independently.
- Extend the dashboard with a BMI-banding visual (e.g., underweight / normal / overweight / obese) to make the BMI–cost relationship easier to read alongside smoking status.

9. Conclusion

The Medical Insurance Cost Analysis Dashboard provides a clear, filterable view of how region, gender, family size, age, BMI and smoking status relate to insurance charges. Across the three segments reviewed, average costs ranged from 9.97K to 14.94K, with maximum charges consistently in the 48K–59K range. The analysis suggests that no single factor drives cost in isolation — smoking, age and BMI interact to produce a small number of high-cost outliers within each segment, which pull the averages upward. Continued monitoring of these segments, supported by the dashboard's slicers, will help identify at-risk customer groups more precisely.